POLITECNICO MILANO 1863 | DIPARTIMENTO DI SCIENZE E TECNOLOGIE AEROSPAZIALI

Combustion in Thermochemical Propulsion: Introduction to the Course

(v01a)

Prof. Christian Paravan

christian.paravan@polimi.it

Eng.. Federico Giambelli

federico.giambelli@polimi.it

Abstract

This presentation provides an introduction to the Combustion in Thermichemical Propulsion (CTP) course.

First, the course objective is detailed. Then, general outline of the lectures is presented. Overall organization of the classes is given, before passing to definition of the final exam.

In the presentation, differences between the two editions of the course (6 and 8 ECTS/CFU) are highligted.

Teaching Staff

Prof. Christian Paravan
christian.paravan@polimi.it
(CP)
Eng. Federico Giambelli
Federico.giambelli@polimi.it
(FG)

1. Objective

2. Lecture Schedule

3. Course Outline
    a. Course Program
    b. References

4. Exam
    a. Written Test
    b. Reports

Objective

01

Provide an introduction to combustion phenomena and their applications to propulsion systems.

The course has a focus on rocket propulsion, though it is not limited to the aerospace field.

Students will achieve knowledge of

i. fundamentals of combustion processes (chemistry, thermodynamics, fluid-mechanics)
ii. advanced topics related to thermochemical propulsion for aerospace applications
iii. engineering methods and notions for the understanding (and implementation) of combustion system design.

Lecture Schedule

02

Mon, 10.15-12.15, L.02
Wed, 10.15-12.15, B2.2.12
Thu, 08.15-10.15, L.03

Lecture will typically start according to the schedule.
It is possible that the actual Lecture will start 15 minutes later than scheduled, in case of short questions from the students (unless transfer time for the class requires a different choice).

Please, understand that if some of the questions require a wide/long answer, this will be moved to a private discussion (with an appointment to be set by email).

The course has no definite discrimination between conventional Lectures and other Activities.

Relevant details on the schedule will be given as early as possible, by WeBeeP. Should no information be available, this means that conventional lectures will take place.

Conventional lectures: CP in class/streaming in the personal Webex room
Other activities: FG in class/streaming in the personal Webex room
(Unless exceptions apply, these will be notified by WeBeep).

Lecture recordings are made available (with a possible short and variable delay).

It is possible that recordings are taken from previous editions of the course
(when contents are unchanged with the current edition and the more recent recording exhibit some technical problems).

Course Outline

03

Part 1: Combustion Fundamentals

1.1 Fuels, oxidizers and propellants
1.2 Phenomenological introduction to combustion processes
1.3 Fundamentals of chemical thermodynamics
1.4 Fundamentals of chemical kinetics in combustion processes
1.5 Governing equations in multicomponent reacting systems
1.6 Laminar premixed and diffusion flames
1.7 Turbulent premixed and diffusion flames
1.8 Igniton and extinction phenomena
1.9 Two phase flow combustion: droplet and spray combustion
1.10 Supersonic combustion and detonation fast than the speed of sound vs deflagration (lower than the speed of light)
1.11 Environmental impact of combustion processes

Part 2: Combustion in Rocket Propulsion, Energetic Materials

2.1 Reactivity and combustion of metals
2.2 Combustion in rocket engines (solid, liquid and hybrid rockets)

The course is available as a 6/8 ECTS/CFU teaching.

Topics that are not required by the 6 CFU edition are marked in the slides by a **bold, light blue sentence: No 6 CFU**.

Available Materials

Mandatory references include:

- Slides discussed during the lectures/other activities
- Short traces of the exercises (when available)

Suggested references include (as a supporting material, note this is not a *must-have*):

- Stephen R. Turns, *An Introduction to Combustion: Concepts and Applications* - 3rd Edition, McGraw-Hill, 2011.

Other sources of information include (**these textbooks exceed the course contents**!)

- Chung Law, *Combustion Physics,* Cambridge, 2006.
- Ken K. Kuo, *Principles of Combustion, Second Edition*, Wiley, 2005.
- Jurgen Warnatz et al., *Combustion – Physical and Chemical Fundamentals, Modelling and Simulation, Experiments, Pollutant Formation*, Springer, 2006.

Some open-literature reference will be suggested (these materials are not mandatory for the exam, being optional). While some article will be recommended by the teaching staff, students could refer to the following publications for cutting ede research works on combustion and energetic materials: *Journal of Propulsion and Power*, *Propellants, Explosives, Pyrotechnics*, *Combustion, Explosion and Shock Waves*, *International Journal on Energetic Materials and Chemical Propulsion, Aerospace.*

Exam

04

Exam: Organization and Details (1 of 4)

Students presenting their subscription once the deadline is passed are not enabled to take the exam.
Thus: subscribe to the exam by the deadline!

Exam is a written (closed book) test with
- (open/closed) questions;
- short exercises.

Questions and exercises are based on the material discussed during classes. The simplest way to understand the possible contents of exam questions is ***if it is in the slide, it is a possible question***. Content and structure of the written text will vary between different exams (e.g., a different number of questions is possible between exams). In general, the exam duration is between 120 and 150 minutes.

Oral discussion is performed only for excellence rating (starting evaluation in the written part > 26/30). **Oral discussions are required from the teaching staff, not from students**. The oral is an option offered to the student that can refuse the oral with no changes on their rating. Oral exam can imply a decreased evaluation with respect to the written part.

No oral exams are taken to reach 18/30, or for marks equal or lower than 26/30.

Exam: Organization and Details (2 of 4)

The final evaluation of the exam combines the mark of the written test (see slide 13) with that of one/two **report/reports**. **These reports are mandatory,** and are related to advanced exercises deepening the topics introduced in the lectures. Report activities are planned for **groups of students**. **Groups are formed and managed by the students**. **Typical (suggested) group** composition includes **5 members**. The mark is the same for all the group members.
The groups will work on selected topics provided by the teaching staff. Topics will be introduced through the classes, with additional material provided with the assignment of the exercise.
The **reports are two (2) for the 8 ECTS/CFU edition of the course, one (1) for the 6 CFU edition. Reports will be based on the analysis of experimental data from combustion tests performed by the SPLab research team**. The **experimental results will remain a property of SPLab, together with the publishing and presentation rights (i.e., no LinkedIn/other social media posts on the results/experiments, no use of the experiments outside of the course by the students).**

The final output of these activities will be a **report (one per each assignment).** The report structure is clarified in the next slide.

The **report(s) must be delivered at least eleven (11) days before the day of the exam** students want ot take part in (i.e., if a student want to take part to an **exam planned for 20 Jun, reports must be delivered by 09 Jun @ 23.59**). Reports will be rated **-1 to 3** points each. The report evaluation will be added to the written exam results of sufficient students (i.e., the **report(s) does(do) not contribute to the 18/30 threshold!**).
Reports will be evaluated considering: (i) correctness of the technical content (implemented methods, results and analyses), (ii) order and clarity of the writing and of the overall document, (iii) clarity and presentation of the figures/tables, (iv) correct use of the references. 6 PAGINE !!! SELEZIONA LE REFERENCES PIU' INFLUENTI

Within the scientific community the use of artificial intelligence (AI) is becoming more and more popular. Yet, AI should not be used to create entire documents, to comment results. Use of AI in the reports must be declared by the authors, clarifying: (i) what AI service was used (e.g., *ChatGPT - Version 5.2 Thinking, Claude – Free,* …), (ii) what AI was used for (e.g., text polishing, reference format check, debugging of original codes), (iii) that the report technical content is due to the authors and not by AI (should AI be used to check the results, this must be clearly stated).

Exam: Organization and Details (3 of 4)

Each report will be limited to 6 pages, in article format. Single/double column formats will be accepted. Font size must be (at least) 11 pts. Article templates from international technical journals as the *Journal of Propulsion and Power,* or *Acta Astronautica*, can be taken as starting schemes to support report's writing (in particular, for the formatting of references and the creation of figures/tables). See Preparation of Papers for AIAA Technical Journals - Overleaf, Online LaTeX Editor, LaTeX instructions for authors| Elsevier.

Students will deliver the report (**1 copy per report per group**) in a proper **WeBeeP repository** (please note a different folder will be available based on the exam date – i.e., a folder will refer to the first call and a different folder will be created for the second call, and so on). Details for the delivery (file name, accepted formats and other) will be given in the description of the delivery folders.

In general, .pdf files will be required for the text, while .m and .xlsx files will be accepted for the data analysis (should other formats be used, these must be freely available – e.g., Python).

What should be included in the delivery?

1. **The report(s) – see next slide for the expected content/structure;**
2. **Data the report is build by (i.e., starting from raw data, students will work in Matlab/Octave/Excel/… to produce graphs and using the given information as inputs for further calculations);**
3. **Figures included in the report(s) will be available as .fig (with the code generating them)/.xlsx (to verify the link originating them)/Other modifiable extension (i.e., all the calculations/all the figure creation must be accessible to the teaching staff)**
4. **An Excel file with the list of the report authors (alphabetical order) with their id. numbers (both person code and matricula).**

Exam: Organization and Details (4 of 4)

The **report is structured as a scientific article, it must include**:

- a list of the authors of the report (with personal code/matricula and email of each member)
- an abstract summarizing the contents and highlighting the achieved results
- a brief discussion of the state of the art of the research
- an introduction of the materials and of the methods used in the analysis (this includes a discussion on data reduction and uncertainty quantification);
- a characterization of the tested materials with an analysis of the results;
- conclusions of the work (a short summary of what was done and of the main outcomes) and future developments (a list of two-three points of further investigation starting from the outcomes of the study).

In the whole report, **references from the open literature are considered (and cited, in proper order)**. In the report **Figures and Tables are inserted and numbered. Each figure/table is cited** in the text **following the numbering order** (the first figure to be cited is the *Figure 1*, followed by the *Figure 2*). Figures and Tables are created to be readable considering the use of A4 as format for the report (thus ticks, font size and line thickness must be properly selected). Each Figure/Table must have a caption describing it.

Topics of the reports will be defined **by mid April**.

POLITECNICO MILANO 1863 | DIPARTIMENTO DI SCIENZE E TECNOLOGIE AEROSPAZIALI

Combustion in Thermochemical Propulsion: Introduction to the Course

(v01a)

Prof. Christian Paravan
christian.paravan@polimi.it

Eng.. Federico Giambelli
federico.giambelli@polimi.it